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Beyond Clearing

Central Counterparties' Investment and Risk Management of Cash from Margin Calls

Elisa Letizia, Pierpaolo Grippa, Franck Viollet

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Beyond Clearing: Central Counterparties' Investment and Risk Management of Cash from Margin Calls

Prepared by Elisa Letizia, Pierpaolo Grippa, Franck Viollet

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ABSTRACT: Central counterparties (CCPs) have become pivotal to financial stability in the post-Global-Financial-Crisis era by mitigating contagion in interconnected markets through the management of counterparty credit risk—the risk that the opposing party in a financial transaction fails to deliver payments or assets. Traditional risk assessments of CCPs primarily address the impact of member or client defaults on CCP liquidity and solvency. This paper introduces a framework that expands this focus to include investment risks associated with the significant cash CCPs hold as margin and applies it to CCPs operating in – or of systematic importance for – the European Union (EU). Specifically, we analyze 'non-default' losses stemming from market risk and counterparty credit risk within CCPs' cash investment strategies. Our findings show that the impact of market risk is generally contained due to strict eligibility criteria and the short duration of investments. However, counterparty credit risk from reverse repo exposures can be substantial, with frequently insufficient haircuts that do not provide sufficient protection against collateral devaluation, posing the potential to significantly deplete available CCP resources. Quantifying these risks sheds light on how margin calls can transmit financial stress from capital markets into money markets via CCPs' cash management practices.

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Beyond Clearing

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Prepared by Elisa Letizia, Pierpaolo Grippa, Franck Viollet [1]

[1] "The author(s) would like to thank" Martin Ockler (ESRB) for sharing CCP data.

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Introduction

Regulatory reforms introduced after the Global Financial Crisis (GFC) have made central counterparties (CCPs) core nodes of modern market structure. Before the crisis, the majority of transactions were concluded bilaterally, so directly between counterparties, with each party being exposed to counterparty credit risk, i.e. the risk that the other party would not deliver payments or assets. The GFC revealed that shocks could be significantly amplified through these opaque interconnected exposures, prompting regulators to require central clearing via CCPs, which act as intermediaries between buyers and sellers (FSB, 2010). CCPs mitigate counterparty risks through mechanisms such as multilateral netting, margining, and default funds. However, this centralization of risk also created a small number of highly interconnected systemic CCPs. Consequently, while CCPs are designed to absorb shocks, their distress could instead propagate and amplify systemic stress. This dual aspect of their systemic importance and centrality in financial systems has led to concerns that CCPs have become 'too important to fail' (Wendt, 2015).

Much of the policy and academic focus so far has centered on default management and the loss-absorbing 'waterfall' that protects CCPs when a clearing member fails. These days CCPs run regular credit and liquidity stress tests to size financial resources against extreme-but-plausible market moves and member defaults. Regulators have increasingly complemented these firm-level exercises with supervisory stress tests that standardize scenarios, compare vulnerabilities across CCPs, and explicitly analyze interdependencies among CCPs. Prominent public implementations of the latter include ESMA's EU-wide CCP stress.

In parallel, academic and policy research emphasizes that the more central role assigned to CCPs in the post-GFC world has led to a partial transformation of counterparty risk into liquidity risk, and that this requires refocusing the risk analysis at macro level from the solvency of CCPs and clearing members to their liquidity (Cont 2017); frameworks have also been proposed for macro-prudential CCP stress testing, while highlighting the need to expand their scope beyond CCPs by identifying and measuring system-wide vulnerabilities across CCPs and their clearing members (Anderson et al 2018) —the most prominent example of this more comprehensive type of exercise being, to this day, the Bank of England's System-Wide Exploratory Scenario (BoE 2024). Much less attention has been paid so far to non-default losses (NDLs)—losses that do not originate from a clearing member default.

This paper focuses on NDLs stemming from the investment of cash resources held by CCPs. While NDLs can also arise from operational sources (for example, cyber incidents or litigation), investment-related losses are the dominant component. CCPs routinely collect large cash balances from their clearing members via margin calls—initial and variation margin—and default fund contributions. This cash is not held idle: CCPs reinvest it in high-quality securities, bank deposits, and reverse repos. CCPs primarily aim to ensure the liquidity of such investments as they need to be ready to be liquidated without material depreciation in the event of a member defaulting (to cover its losses). They also seek an adequate return on the resources they receive from their members, while minimizing the opportunity costs of potentially large liquidity holdings.

NDLs from investment arise from market losses on securities and counterparty credit losses on deposits and reverse repos. These losses can rapidly erode equity and liquidity and undermine confidence even without losses from members' default. Quantifying NDLs helps authorities and CCPs complement their assessment of the adequacy of existing governance, capital, insurance, and recovery tools to keep critical

clearing services running in stress. It can also help avoid a situation where a CCP that could have survived members' default fails because it incurs additional non-default losses from its investment and necessitates disruptive recovery actions (e.g., haircutting gains, service restrictions) with broader financial-stability spillovers. An initial attempt to estimate the size of these losses—for CCP's investments in bonds—can be found in ESMA's 5th Stress Test Exercise for Central Counterparties (ESMA 2024b).

We developed a framework to quantify potential losses of such investments accounting for both losses from market risk and counterparty credit risk and applied it to CCPs operating in the EU—or categorized as systematically important for the EU. The framework is based on historical simulations (such as those used for historical-based estimates of Value-at-Risk), which have two advantages: on one hand, they overcome the limitations of testing losses under specific scenarios, which might not exhaustively cover the probability space for extreme events; on the other hand, by not assuming a particular distribution of the asset returns, the setup does not rely on specific model for market dynamics, allowing to cover, in a coherent manner, the several market factors to which CCP might be exposed. For reverse repo positions, we further combine market dynamics with simulations of the defaults of repo counterparts, to capture counterparty credit risk.

When applying the framework to the cash investments of CCPs authorized to operate in the European Union, we find that counterparty credit risk matters more than market risk. Losses from market risk on direct investments are generally contained, reflecting strict eligibility and maturity constraints over their investable assets. By contrast, counterparty credit risk embedded in reverse-repo exposures can be sizable for a subset of CCPs and materially affect their capital positions. These estimates provide a practical input for financial-stability monitoring and for calibrating recovery and resolution triggers[2] that explicitly account for investment-related NDLs.

One driver of notable counterparty credit losses is the low (if not zero or negative) level of haircuts for collateral observed in many CCPs' bilateral repo trades. While this is a prevalent feature of bilateral repo markets across various jurisdictions, it entails that even a moderate asset depreciation can leave the cash leg of the repo (in this case that of the CCP) with insufficient collateral, in case of counterpart's default. From a risk management perspective, the appropriate level of haircut should reflect the level of risk of the two counterparts entering the repo trade.

We therefore additionally investigate the appropriateness of existing collateral haircuts. We run a simple 'symmetric' haircut exercise that uses counterparties' default probabilities and collateral tail risks to assess when non-positive haircuts could be justified from a risk-management perspective. Under plausible assumptions on a CCP's rating (available, in reality, only for few), we find that zero or negative haircuts are rarely—if ever—justified from a risk management perspective. Any deviations from the "ideal" level of haircut would need to be covered by an appropriate level of capital dedicated to such exposures.

The rest of the paper is organized as follows. Section 2 reviews CCP cash-management practices and the EU regulatory setup. Section 3 presents the framework for estimating investment-related NDLs for direct investments and reverse repos. Section 4 discusses haircut calibration and the conditions under which zero or negative haircuts might be risk-justified, together with capital implications. Section 5 concludes.

[2] FSB (2024a).

Background

CCPs are expected to manage financial risks not only in their clearing activity, but also in non-clearing activities, including the investment of cash resources. For the cleared part of their business, CCPs rely on the default waterfall.[3] A CCP default waterfall is a pre-defined sequence of financial resources designed to absorb losses arising from a clearing member default. The waterfall starts with the defaulting member's margin and default fund (DF) contribution, exhausting the defaulter's resources before impacting other participants. If these resources prove insufficient, the CCP makes a limited (capped) contribution to offset the remaining loss: this contribution is sometimes referred to as "skin-in-the-game" (SITG) as it reinforces prudent risk management incentives. Subsequently, the mutualized default fund, comprised of contributions from non-defaulting members, is utilized to absorb further losses. In recovery scenarios, the CCP may invoke assessment rights to call additional resources or employ recovery tools such as variation margin gains haircutting or partial contract tear-up. The structured hierarchy of the waterfall is designed to ensure loss absorption while preserving market confidence and continuity of critical clearing services.

For liquidity management, the CPMI–IOSCO Principles for Financial Market Infrastructures (PFMI) set strict requirements for how CCPs should safeguard and invest both their own capital and participants' assets. Investment policies should prioritize safety and liquidity over profit and focus on instruments secured by—or representing claims on—high-quality obligors that can be liquidated rapidly with minimal price impact. Consequently, investments are restricted to instruments with negligible credit, market, and liquidity risks (CPMI-IOSCO 2012, Principle 16). Furthermore, CCPs are expected to actively monitor concentration risks and diversify their custodians whenever feasible. The rest of this section describes the CCP' investment principles, with implications on asset allocations in various jurisdictions and profit generation, and the interplay with capital requirements and recovery and resolution.

EU CCPs: market and regulatory framework

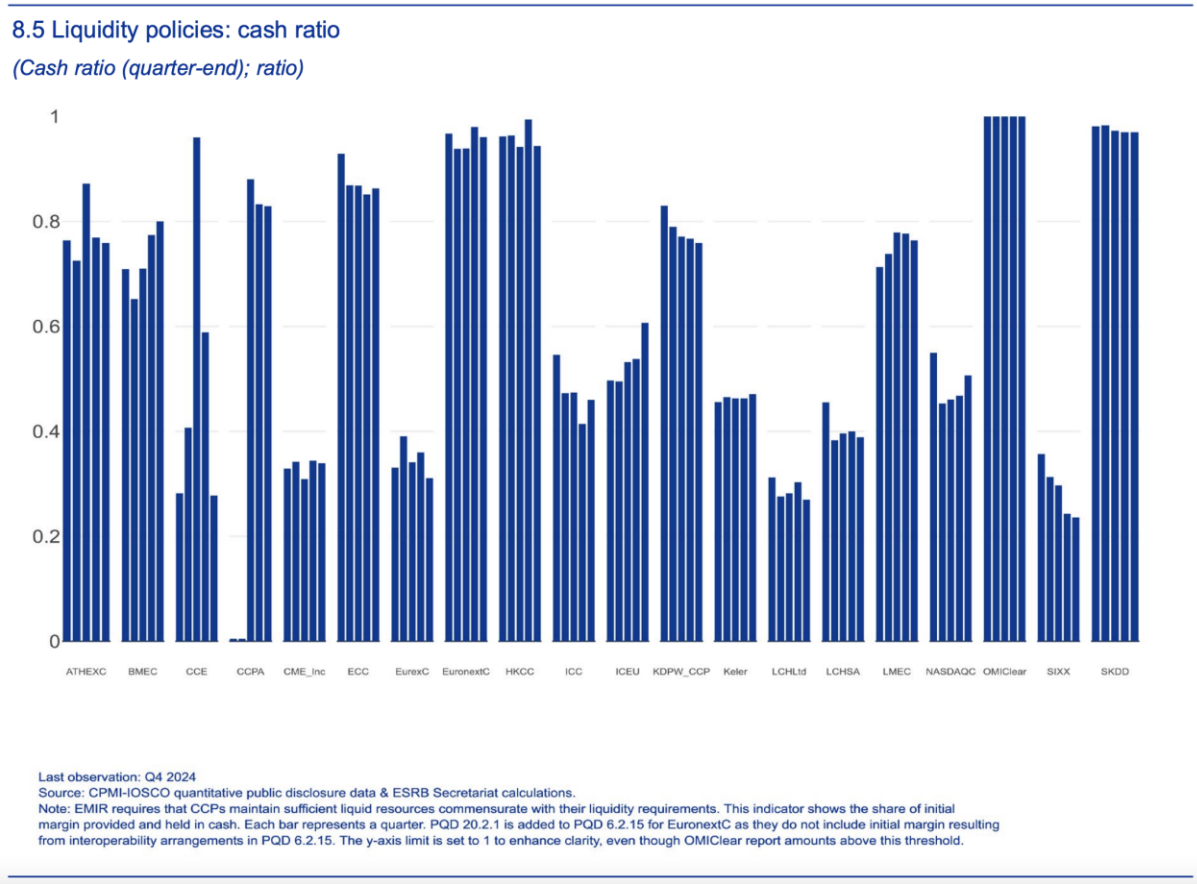
There are 14 CCPs in the EU across 12 member states, plus two Tier 2 CCPs in the UK.[4] In the EU, Article 47 of EMIR sets general principles for managing CCP investment risk, and the related regulatory technical standards (RTS 153/2013) specify detailed risk-management requirements. EU CCPs invest a sizable share of their liquidity—more than 20 percent on average—in reverse repos (Aldasoro et al. 2023, Graph 4. A), typically backed by highly rated sovereign collateral.

A material share of pre-funded resources in the waterfall is collected in cash (Figure 1). Pre-funded resources are mainly initial margin (required and excess) and default fund contributions: for example, in the EU, required margin stood at EUR 612 billion as of March 2023, while the default fund totaled EUR 35.2 billion. Skin-in-the-game and other pre-funded resources are much smaller (EUR 0.8 billion). Pre-funded resources exclude routine variation margin (VM), which is typically settled by the end-of-day margin cycle (CPMI–IOSCO 2025). Clearing members are usually required to post a minimum share of initial margin in cash. As of March 2023, collateral held by in-EU and Tier 2 CCPs consisted primarily of government securities (48 percent) and cash (45 percent) (ESMA 2024).

[3] For example in the EU, this is described in Regulation (EU) No 648/2012 (EMIR). Article 45. Default waterfall

[4] Tier 2 CCPs are non-EU CCPs identified as systemically important for EU financial stability.

Figure 1. Cash Ratio for IM Received—Sample of EU CCPs (ESRB 2025)[5]



When volatility rises or during crises, market participants tend to increase the cash share of posted collateral, prioritizing liquidity over cheapest-to-deliver securities. [6] As volatility subsides, they tend to progressively reduce the proportion of cash while optimizing collateral. Such cash is typically invested to ensure an adequate return to both the CCP and its members (Figure).[7] CCPs usually reinvest received cash to generate treasury income. While modest in relative terms, this may constitute a significant source of income. In normal conditions – top panel (a) – a CCP's investment profit would constitute the "Treasury income", part of which is distributed to members (Treasury expense). The difference, which is typically a flat "Management fee",

5 The chart covers EU-authorized CCPs and third-country CCPs reporting under the CPMI-IOSCO PQD framework. Acronyms: ATHEXC (Athens Exchange Clearing House, EL), BMEC (BME Clearing, ES), CCE (Cboe Clear Europe, NL — formerly EuroCCP), CCPA (CCP Austria, AT), ECC (European Commodity Clearing, DE), EurexC (Eurex Clearing, DE), EuronextC (Euronext Clearing, IT — formerly CC&G), KDPW_CCP (PL), Keler (KELER CCP, HU), LCHSA (LCH SA, FR), NASDAQC (Nasdaq Clearing, SE), OMIClear (PT), SKDD (SKDD-CCP Smart Clear, HR). Third-country CCPs: CME_Inc (US), HKCC (HK), ICC (ICE Clear Credit, US), ICEU (ICE Clear Europe, UK — Tier 2), LCHLtd (LCH Limited, UK — Tier 2), LMEC (LME Clear, UK), SIXX (SIX x-clear, CH)

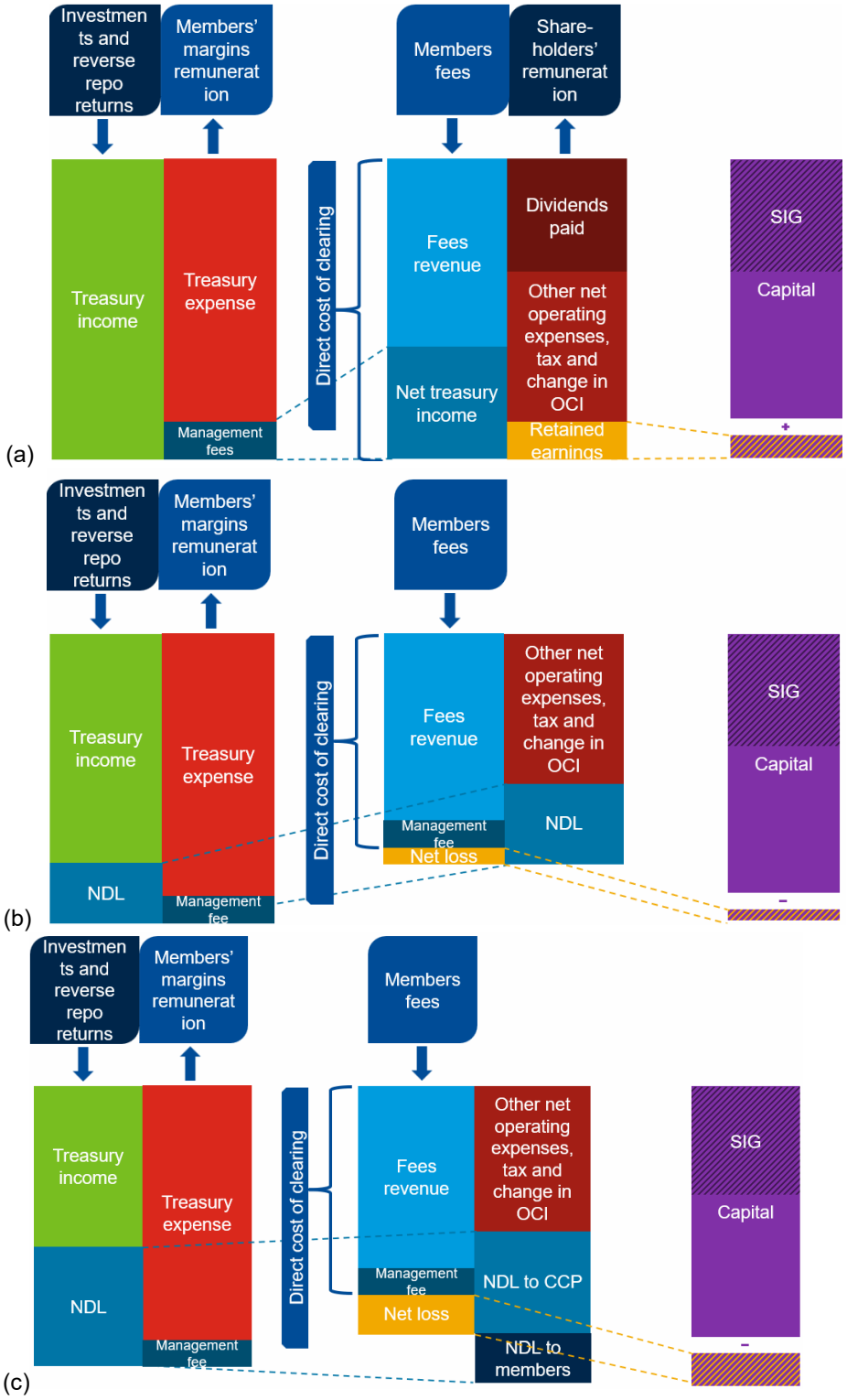
6 Mobilising collateral key to avoiding a liquidity crisis, clearing experts say | FIA

7 FIA and ISDA (2020)

remains in the CCP in the form of “Net treasury Income.” While modest in relative terms, this may constitute a significant source of income for the CCP and, together with the fees for the cleared business, represent the direct cost of clearing for the members. If such income exceeds operating costs, profits are either paid to shareholders or kept as retained earnings to build capital. In case of negative market return on investments—middle panel (b) and bottom panel (c)—the CCPs are likely to still charge their members a management fee, but they must decide how to absorb the losses from the investment portfolio (NDL). One option is to use own resources—panel (b)—for example by curtailing dividends for the year and if not sufficient deducting from own capital.[8] Alternatively—panel (c) —the CCP can trigger its recovery plan, which often includes a threshold for NDL absorption, with the remaining losses being distributed to members.

[8] This was the case of Cassa di Compensazione e Garanzia S.p.A. in 2022: when faced with losses on its direct investments due to raising interest rates, it cut sharply its dividends for that year, with no impact on capital.

Figure 2. CCPs' Investment Income and Profit and Losses



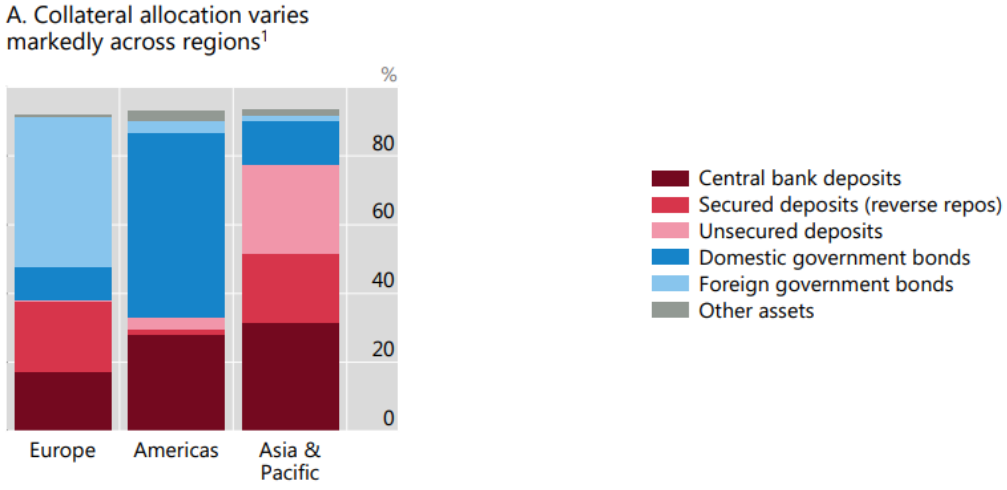
Source: authors

CCPs' investments principles

CCP investments must be quickly liquidable and with minimal adverse price impact. In the EU, Article 47 of EMIR and RTS 153/2013 govern CCP investments, mandating that assets must be highly liquid, diversified, and minimally exposed to market or credit risk to ensure prompt availability of funds with negligible adverse price impact. CCPs must adopt prudent investment policies that preserve principal, typically restricting holdings to cash or highly liquid, low-risk instruments (such as government or central-bank-guaranteed debt), while capping the average portfolio time-to-maturity at two years. Cash not deposited at a central bank must be placed with authorized, highly secure financial institutions, with at least 95% collateralized by highly liquid assets. . In ESMA's fifth stress-test, data show CCPs held EUR 164.7 billion in secured cash via uncleared reverse repos, predominantly in USD (47 percent), EUR (34 percent), and GBP (17 percent) (ESMA 2024b).[9] Additionally, collateral assets must be fully protected, preferably through recognized securities settlement systems, while CCPs must also ensure full transparency of their investment policy and submit it to ongoing regulatory oversight.

CCP cash allocation varies across jurisdictions (Figure 3). Consistent with prudent investment principles, CCPs primarily invest their cash in central bank deposits, bank deposits, and government securities, though the mix differs. European CCPs rely more on reverse repos than US and Asian CCPs and make less use of central bank deposits. Differences partly reflect different access conditions to central bank facilities (Table 1). EU CCPs, given their limited access to Federal Reserve facilities, use other arrangements for their USD deposits. In Europe, the relatively lower volume of central bank deposits can be explained by the large presence of cross-border business: for example, while EA CCPs can deposit their EUR cash at the central bank and receive market remuneration (Euro short-term rate - 20 bps), a non-EA (or non-EU) CCP authorized to clear in the EU can typically allocate their EUR deposit, but without remuneration.

Figure 3. CCPs' collateral allocation, by geographical area



Source: Aldasoro et al. (2023)

[9] For example, based on their public disclosures for the 3rd quarter of 2025, EUREX Clearing (which operates also with a banking license) deposited almost 93 percent of its EUR liquidity at the ECB, while LCH Ltd (based in the UK and with no direct access to ECB liquidity) held 65.6 percent of its EUR liquidity in the 'secured bank' category, which includes reverse repos.

Table 1. CCP access to central bank facilities

	Euro area	*Japan*	*US*	*UK*
Deposit account	Remunerated for Authorized / supervised euro area CCPs;[10] Non remunerated for non-EA authorized CCPs	Regulated CCPs	Eight systemically important domestic CCPs	Regulated CCPs
Lending	Euro area CCPs licensed as credit institutions (Marginal Lending Facility), Authorized / supervised euro area CCPs with a dedicated overnight credit facility[11] withinTARGET[12]	No	No	Regulated CCPs

Source: Authors

[10] ECB confirms remuneration ceiling for euro area government deposits and adjusts remuneration of other non-monetary policy deposits; Decision (EU) 2024/1209 of the European Central Bank of 16 April 2024 on the remuneration of non-monetary policy deposits held with national central banks and the European Central Bank (ECB/2024/11)

[11] ECB Press Release: ECB introduces changes to the dedicated credit facility for euro area CCPs

[12] TARGET (Trans-European Automated Real-time Gross settlement Express Transfer system) is the Eurosystem's real-time gross settlement (RTGS) infrastructure for the euro, operated by the ECB and the euro area NCBs. It settles large-value and time-critical payments in central bank money on a transaction-by-transaction basis, with immediate finality.

Liquidity management

CCPs should maintain sufficient liquid resources to meet obligations under both idiosyncratic and market-wide liquidity stress. They can raise cash through several channels:

- **Maturing Investments:** Cash collateral from margins and default fund contributions, typically invested on a short-term basis, can be retained upon maturity rather than being reinvested.
- **Repo Transactions and FX credit lines:** High-credit quality assets held in the investment portfolio or assets received through settlement fails/defaults can be used as collateral in repo transactions and FX credit lines to raise cash. This provides same-day liquidity with maturities tailored to the specific shortfall period. For reverse repo investments, the CCP may also retain the right of rehypothecation.
- **FX Transactions:** In case of excess liquidity in one currency, the CCP can execute short-term FX funding swaps to fill a shortfall in another currency.
- **Asset Liquidation:** The CCP may sell portfolio assets to market dealers at the earliest possible value date.

Liquidity planning should also account for concentration costs—additional liquidation expenses beyond the initial market shock that CCPs would face when liquidating (through auction or hedging) concentrated positions. Such costs may arise when liquidating defaulting clearing members' portfolios or collateral received in reverse repos with a defaulting counterparty. Where these positions are material, a CCP would likely need to accept a discount on their market value in order to liquidate them. Unlike the impact of a prescribed market stress scenario, which for a given portfolio can be calculated with precision, concentration costs are not directly observable: CCPs and regulators estimate them using models, dealer surveys, or stress-test assumptions.

In practice, CCPs and regulators use several approaches to estimate concentration costs, each requiring calibration and subject to uncertainty. For example, concentration costs may be estimated using theoretical models; or alternatively, via dealer surveys: while these can incorporate market expertise, they depend on sufficiently granular and consistent inputs. ESMA's fifth CCP stress test assumed that liquidating investment-grade sovereign bonds with 1 to 5-year maturity would generate losses of 15 bps (respectively, 30 bps) when positions represent 50 percent (respectively, 100 percent) of average daily volume (ESMA 2024b; Table 2).

Table 2: IG Sovereign Bond liquidation costs (in bps) given the position value as a portion of average daily volume

	Position value as share of the Average Daily Volume			
	25%	50%	100%	200%
Short-Term (< 1 yr)	7	14	26	51
Medium-Term (1–5 yrs)	8	15	30	61
Long-Term (> 5 yrs)	14	29	56	108

Source: ESMA

Market risk and counterparty credit risk management and regulation

CCPs mitigate investment risks primarily through conservative eligibility criteria and quantitative limits, while disclosures provide less information on the dynamic measurement and monitoring of market and counterparty credit risk (CPMI-IOSCO 2023). In addition to holding highly liquid assets, CCPs commonly apply rating and maturity limits to the debt instruments they hold, as well as concentration limits by issuer, country, currency, and counterparty. Public disclosures document the adoption of sensitivity analysis, stress testing, and risk models (VaR/ES); however, most of these refer to the CCPs' risk management activity in the clearing space, with limited detail on the investment of cash from margin calls**.**

ESMA has translated the investment-policy principles of Article 47 of EMIR into more detailed prudential rules. RTS 152/2013 sets capital requirements for CCPs, including for credit, counterparty, and market risk. The rules for the calculation of these requirements are borrowed from the bank capital regulation framework (CRR). For market risk, CCPs can choose between the Standardized Approach and the Internal Models Approach, while for counterparty credit risk on securities financing transactions (including reverse repos), they can choose between the Mark-to-market Method (art. 274 CRR) and the Financial Collateral Comprehensive Method (Art. 223-227 CRR).

Information on the application of the capital requirements for non-clearing activities is generally available only for CCPs with a banking license, which are directly subject to banking disclosure rules. Where disclosures exist, CCPs often use the Standardized Approach for market risk and the Financial Collateral Comprehensive Method for counterparty credit risk. For CCPs without a banking license, public information is typically limited, including on the level of capital requirements.

CCP recovery and resolution

Recovery and resolution are distinct phases in managing CCP distress and potential failure, each with a specific role in safeguarding financial stability. Recovery comprises CCP-led measures to restore viability using pre-agreed tools and procedures outlined in its recovery plan, when facing a severe financial distress scenario with losses exceeding those covered by standard default management procedures (i.e., when the default fund is exhausted). Resolution begins when recovery is insufficient or would threaten broader financial stability, and is led by a resolution authority under a dedicated legal framework.

Recovery tools—such as variation margin gains haircutting, cash calls, or partial tear-up of contracts—aim to allocate losses and restore a matched book without resorting to public support, while maintaining continuity of critical clearing services. Resolution steps in when recovery ends: in case a CCP cannot continue critical operations through its own efforts, the resolution authority takes over, aiming to maintain the CCP's critical operations, for example through loss-sharing arrangements or a transfer of its contracts to a different clearing house. As set out by the Financial Stability Board (FSB 2025), the objective of CCP resolution should be the pursuit of financial stability and ensuring the continuity of critical CCP functions in all jurisdictions where those functions are critical, without exposing taxpayers to the risk of loss.

Resolution is initiated when the recovery plan is either insufficient to restore viability or its implementation would compromise broader financial stability. Resolution is undertaken by a designated resolution authority under a legal framework, such as the EU's CCP Recovery and Resolution Regulation

(CCPRRR). The aim is to ensure the continuity of critical functions while minimizing disruption to financial markets and avoiding systemic contagion. Tools available in resolution may include the transfer of critical services, the write-down or conversion of liabilities, and the use of a bridge CCP.

CCP recovery plans—prepared ex ante and approved by authorities—must cover a comprehensive range of extreme, forward-looking scenarios. Resolution planning by the authorities similarly considers a reasonable range of default loss scenarios, non-default loss scenarios, and combinations of both. This paper focuses on NDLs arising from investments—for example, if an reverse repo counterparty fails. A CCP may need to bear such losses, in the absence of other loss allocation tools in its own rulebook. Losses could also occur from the investment of a CCP's own financial resources.

Quantifying non-default losses (NDLs) supports the ongoing effort of designing credible recovery and resolution strategies for CCPs. In a downturn, NDLs can add to spillovers from CCPs to clearing members and the broader market ecosystem. Better estimates of NDLs can help calibrate recovery and resolution triggers, as well as inform whether CCP own resources are sufficient if those triggers are reached, improving the credibility of plans under adverse conditions.

Estimating Non-Default Losses

CCPs operating in the EU invest idle liquidity in securities and reverse repos, with considerable heterogeneity across institutions. Securities investments generally preserve currency composition and add little incremental FX risk. Weighted-average maturities are below 1.5 years (average about 0.7), and durations are typically short—often below three months—though a few CCPs hold longer-duration portfolios.

Reverse-repo usage varies widely. Of the 14 EU CCPs, seven use reverse repos, mostly in non-EUR currencies (USD, GBP, and other EU currencies). Two non-EU Tier 2 CCPs allocate more than 87 percent of their EUR cash margins to reverse repos. EU CCPs typically face a small set of counterparties (up to eight), while Tier 2 CCPs may face many more (up to 54). Counterparties are usually large financial institutions and often clearing members, though some non-member counterparties appear in the data.

Reverse-repo transactions are characterized, on average, by low—and sometimes zero or negative—haircuts, consistent with practices in bilateral repo markets. This contrasts with CCPs' collateral haircuts in the clearing space, which are typically much larger for the same securities when posted to meet margin calls. For example, depending on maturity, centrally cleared index-linked gilts have haircuts of between 0.5 and 35 percent at LCH Ltd in December 2025. This lower protection from haircuts in the non-clearing spaces might expose CCPs to losses if a reverse-repo counterparty defaults during a market downturn, when collateral values may fall and liquidation conditions may be strained.

Table 3: Reverse Repo Sample Descriptives Statistics

	EU	Tier2
Use repo	7 of 14	2 of 2
Max number of counterparties	8	54
Average Number of Collateral received	54	199
Average Duration of Collateral received (years)	4	9.7
Coverage (percent)	100	26
Average repo (EUR bln)	9	47
Total repo (EUR bln)	74	94
Average SITG (EUR m)	32	137
Total SITG (EUR m)	253	275

Source: ESMA, CCP Disclosures, authors calculations. Dec 2024

We adopt an Expected Shortfall (ES) approach to estimate the potential unexpected NDL on CCPs' investments. For direct investments, we construct loss distributions via historical simulations over January 2020–December 2024, which allow us to capture model-free co-movements across yield curves, tenors, and FX rates over a five-day horizon. For reverse repos, we combine collateral value shocks from the same historical simulations with simulated defaults of repo counterparties to capture counterparty credit risk. [13] In both estimates, we test the resilience of CCPs without them counting on any resources to cover losses beyond those already freely available to them.[14] Concentration add-ons linked to the cost of liquidating concentrated positions, mentioned above, represent an additional source of risk with respect to that of NDLs, which is not modelled in the simulations.

We benchmark estimated losses against proxies for loss-absorbing capacity. Because investment-related NDLs fall outside the clearing waterfall, in normal times, they should be absorbed by equity, in line with principles of corporate finance. In recovery, most CCPs have provisions in their plans which cap the maximum loss absorbed by the CCP and allocate the remaining losses to members. In the absence of consistent public data on capital requirements for all CCPs,[15] we compare losses to the default fund and to skin-in-the-game (SITG) as practical proxies.

Data inputs vary by instrument. For direct investments, we use portfolio data from ESMA's fifth CCP stress test (reference date 17 March 2023) which was the first application of our framework, with ISIN level granularity. For reverse repos, we combine several sources with reference date December 2024, including the Securities Financing Transactions Regulation (SFTR) transaction data on repos, collateral, and

[13] The use of the 2020-2024 period for historical simulations in modelling counterpart credit risk on reverse repos – a period encompassing severe market stress events like the outbreak of the COVID-19 pandemic and the contagion fears following the collapse of SVB and some regional banks in the US – requires highlighting important caveats: our simulation results depend on historical realizations over a short time period that is not necessarily representative of the co-movement of CCP counterparts' PD over the longer term; in the case of the pandemic, unprecedented levels of policy support may have caused a disconnect between PDs (generally spiking around the stress event) and actual defaults (largely suppressed by the public policy support); results are also contingent on cross-correlations that were specific of the 2020-2024 period and, again, not necessarily indicative of longer-term correlations. On the other hand, the use of NUS-CRI's hybrid market-accounting-based PDs (as opposed to fully market-based ones) might partially mitigate some of these effects, thanks to their lower reactivity to market dynamics.

[14] The logic is the same generally adopted for liquidity stress testing of banks, in which potential liquidity injections by the central bank are not considered, although in historical stress episodes that has often occurred.

[15] Ideally these NDLs should be compared with the CCPs' capital requirements for market and counterparty credit risk. In the absence of publicly available figures on such requirements for all CCPs, we use DF and SITG as alternatives.

counterparties.[16] We combine these with data on SITG and DF from regulatory returns. The SFTR sample available to ESMA only covers trades reported by EU firms: for EU CCPs we rely on own transaction reporting obtaining 100 percent coverage (except for one EU CCP, which we exclude from the sample due to data quality reasons). For Tier 2 CCPs, after cleaning the data for outliers, coverage is partial (with at most 29 percent of reverse repo positions for one Tier2 CCP), so estimated losses are a lower bound. Market time series for relevant yield curves (one for each sovereign in CCPs' portfolios, at different tenors) and FX rates come from commercial data providers, while the probabilities of default (PDs) of CCP reverse repo counterparts are drawn from the NUS-CRI database.[17]

Direct investments

We approximate bond revaluation using duration and convexity. We map each bond to the corresponding bond index, in terms of issuer country and tenor, to determine the relevant time series of yield over the historical simulation horizon.[18] We denote by $V_L(t)$ the value in denomination currency of a bond on any day t of the historical simulation, and $\Delta_k y(t)$ the change in yield over the holding period of length $k = 5$ (i.e. between $t - k$ and t). Given the value of the portfolio at the cut-off date T, we can approximate the change in value over the historical simulation with the following:

$$\Delta_k V(t,T) = V(t) - V(t-k) = V_L(T) \cdot {}^{\%}\Delta_k V(t,T)$$

$${}^{\%}\Delta_k V(t,T) = \frac{\left(FX(t) - FX(t-k)\right)}{FX(t)} - \left(\mathrm{D} \cdot \Delta_k y(t,T) - \frac{s}{2}\Delta_k y(t,T)^2\right)$$

where $FX(t)$ is exchange rate to EUR at time t, D is the duration and s is the convexity at the cut-off date T.

We apply our framework to the eight CCPs with non-negligible direct investments. The median expected shortfall at 1 percent is about 4 bps of invested assets, consistent with short durations. Relative to loss-absorbing

[16] Data provided were based on Trade-State tables. Trade-state is a 'stock' representation of the data, insofar as it captures the state of a transaction at the end of a reporting date. Scope: repo and repo equivalents (i.e., buy-sell-back transactions) where one of the counterparties was in the list of funds in scope. For CCPs, where the trade involved EU CCPs, we considered only the reports from the CCPs, whereas for trades involving non-EU CCPs we had to rely on the reports from European clearing members. Data preparation: The sampling procedure included the pairing of double-sided transactions, and their deduplication – in this regard, in case of double reporting, the quantitative fields contained the average of the information reported (e.g., notional amount, collateral amount). Outstanding repos: for each reference date, we considered outstanding contracts, i.e., repos with a start date at or before the reference date and maturity after the reference date. Cleaning procedure: a range of checks were applied to these data, including outlier identification methodologies, the removal of stale records and records with implausible collateral-to-loan ratios. For more information on the cleaning procedure, see also section Statistical methods in the ESMA EU Securities Financing Transactions markets (2024).

[17] NUS-CRI PD is a forward-looking point-in-time probability of default measure which is produced on a daily basis by the Credit Research Initiative of the National University of Singapore (https://nuscri.org/en/ourdata/). It combines a reduced-form model, based on a forward intensity construction, and a Merton-like structural model, using Distance-To-Default as one of its input covariates. The other covariates include both market-based and accounting-based firm-specific attributes, as well as macro-financial factors (https://d2.nuscri.org/static/pdf/Probability%20of%20Default%20White%20Paper.pdf). In the period considered in this work, the PDs of CCP counterparts ranged from a minimum of 0.017 percent (less than 2 bps) to 5.31 percent, with the peak for the whole sample reached on April 3, 2020.

[18] The change in yield for a given bond is determined by interpolating across the change in yield for the bond indices with tenors closer to its duration.

capacity, the median loss is 1.1 percent of SITG and 1.4 percent of the default fund, with interquartile ranges of roughly 0.5–3 percent and 1–2 percent, respectively.

Reverse repos

For each day in the time series, we compute the potential conditional loss for each CCP-counterpart pair as the difference between the discounted value of the collateral and the cash at maturity, assuming losses can be netted across positions of the same counterparty, but not across different counterparties. Loss given default after the collateral value is assumed to be 1, for lack of information on the treatment of such residual losses. Conditional loss for each reverse repo position is estimated based on the stressed value of collateral and cash legs. For the collateral leg, we use the same approach as for direct investments. For each counterparty i we define the excess collateral C, accounting for the shock on FX on the cash leg in a similar way as for the collateral leg:

$$C_{i,t} = V(T) - cash \cdot FX(T)\left(1 + \frac{FX(t) - FX(t-k)}{FX(t)}\right)$$

For each CCP, we consider all possible combinations of defaults of counterparts and compute the total loss. For CCP counterparts we adopt their (time-varying) PDs from the NUS-CRI database. For a CCP with n_{cpt} counterparties, there are $2^{n_{cpt}}$ combinations of defaulting counterparties. We derive the probability of each default combination, based on individual counterparties' PDs on a given day of the historical simulation.[19] The CCP loss in any given day is the sum of the conditional losses for defaulting counterparties. Denoting with $u^{[k]},\ n_{cpt} \times 1$ the vector of indicators of default for a given combination (i.e. $u^{[k]}{}_i = 1$ if counterparty i defaults), the losses over the historical simulation are given by

$$\mathrm{L_k} = \underbrace{[[C + H \cdot S]^T]^-}_{\substack{conditional\ loss,\\ by\ counterparty,\\ by\ day}} \cdot u^{[k]} \qquad \mathrm{k} = 1, \ldots, 2^{n_{cpt}}$$

where C is a matrix of size $n_{cpt} \times n_{day}$ of excess collateral for each counterparty and day in monetary amount; $H,\ n_{cpt} \times n_{sec}$ is the matrix of collateral posted, in monetary amount in euro at the cutoff date; $S,\ n_{sec} \times n_{day}$, matrix of percentage shocks $\Delta^{\mathrm{percent}}{}_k V(t,T)$. The loss estimate is defined as the expected shortfall at level α across all combinations and all days, assuming each day has equal weight. For our results we set $\alpha = 0.001$.

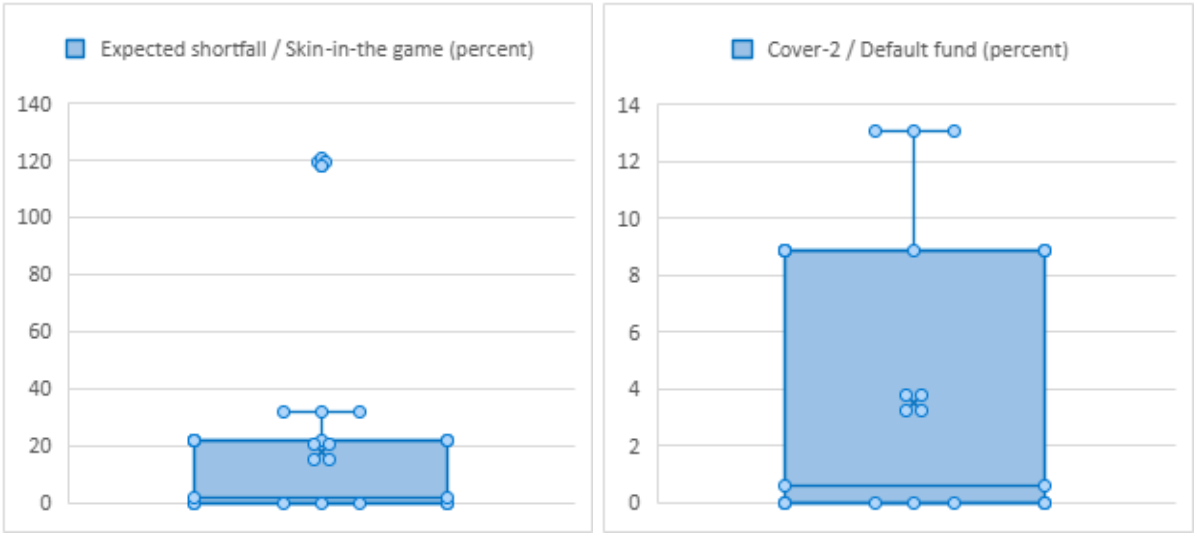
As an alternative metric, we consider the Cover-2 losses. In CCP risk management, cover-2 losses indicate the potential maximum losses from the default of the two clearing members that would cause the largest shortfall of pre-funded resources, and it is used to calibrate the size of the default fund. We translate this concept in our setup by considering the sum of the two largest conditional losses for each day. We define NDLs-Cover-2 as the

[19] By having borrowers' defaults being co-dependent via a common (possibly latent) factor affecting their respective PDs, we capture the systematic component of credit risk; by enumerating all default combinations with their respective probabilities, we also capture most of the idiosyncratic component of credit risk; what is potentially left out of the simulation is the possibility of tail non-linearities that were not experienced in the sample period (an intrinsic limit of the historical simulation approach) and of potential network effects that go beyond what is already embedded in available market and accounting information at the time of PD estimation (like, for example, the increasing interconnectedness between two counterparts caused by the build-up of (net) reciprocal derivative exposures beyond 'usual' levels).

expected shortfall at level α across all days, assuming each day has equal probability. For our results we set $\alpha = 0.001$.

Counterparty credit risk from reverse-repo positions is modest in basis points relative to invested cash but can be large relative to some CCPs' own resources. In most cases, ES losses are small (up to about 15 bps of invested cash), and some CCPs show no losses because stressed collateral values remain above the cash leg. Losses are concentrated in a few CCPs, typically those with relatively small SITG compared to the default fund.[20] When compared to the skin-in-the-game—which we use as a proxy of the amount of losses to be absorbed directly by the CCP—we find that in some cases losses could consume a significant part of own resources, and in one case exceed SITG by around 20 percent, suggesting that a material share of losses from counterparty credit risk could be allocated to clearing members in recovery (Figure 4). Finally, comparing an NDL-style Cover-2 metric to the default fund, we find that NDL-Cover-2 can reach about 13 percent of the default fund, with several CCPs above 7 percent—implying that investment-related NDLs could amplify spillovers in a market downturn.

Figure 4. CCPs' NDLs from reverse-repo exposures



Source: authors

Indeed, while non-default losses (NDLs) typically result in lower aggregate exposures than default losses, their potential magnitude remains material from a CCP resilience perspective. Cover-2 default losses for March 2023 were found to be reaching up to 42 percent (and 49 percent including concentration

[20] According to Huang and Takáts (2020), relatively lower skin-in-the-game can be considered a good predictor for CCP risk appetite.

costs) of the pre-funded resources in the 5th ESMA Stress Test. The median consumption was 10 percent (19 percent with concentration costs).

Collateral haircuts in CCPs' non-centrally cleared repo

Bonds posted as reverse-repo collateral are subject to haircuts, usually calibrated using standard risk measures such as Value at Risk (VaR) or Expected Shortfall (ES).[21] In 2015, the FSB introduced a framework for haircuts on non-centrally cleared securities financing transactions (updated in 2019 and 2020), combining qualitative standards with numerical haircut floors by asset class and maturity (FSB 2020). The framework applies only in specific cases (notably cash financing to non-banks against non-government collateral), which cover only a small share of collateral accepted by CCPs. In practice, haircuts on CCPs' bilateral reverse repos are often below these floors and are sometimes non-positive (i.e. the collateral's current value is not always larger than the 'cash leg' of the repo); in our sample, about 40 percent of transactions have negative haircuts.

Zero or negative haircuts on government bonds are not unusual in bilateral repo transactions. The FSB reports that, based on data from nine G20 countries,[22] approximately 70 percent of non-centrally cleared repo transactions operate with zero haircuts (FSB 2026). Zero haircuts on gilts are documented for the UK repo market, where they represent around 40 percent of bilateral repo volume (Ivan et al 2024). In the US, the Office of Financial Research finds that in the non-centrally cleared bilateral market, "During the first five months of 2025, 56 percent of outstanding repo had zero haircuts while 34 percent and 10 percent had positive and negative haircuts, respectively" (Cenicola et al 2025), while Lu and Wallen (2025) find that in the Treasury repo market dealer banks, on average, lend at negative haircuts; Bejarano and Lu (2026) also find that US dealer banks borrow at positive haircuts from cash lenders and lend at negative haircuts to cash borrowers, deploying their internal funding to supply more leverage to the Treasury market.[23]

For the EU, Hermes et al. (2025) find that about 9 percent of EUR-denominated repos and 4 percent of USD-denominated repos involving euro area entities feature negative haircuts. They discuss three potential explanations: intragroup transactions, 'specialness' premia for specific securities, and market power. For EU CCPs, intragroup motives are unlikely because counterparts are typically unaffiliated financial institutions; 'specialness' also seems less plausible given CCPs' relatively simple investment activity; market power doesn't appear plausible either—at least for large CCPs—but cannot be ruled out completely and merits further investigation.[24]

[21] Sakurai (2025) discusses a number of approaches for the estimation of haircuts for central banks' collateral operation (but applicable in general to all types of counterparts), depending on the collaterals' marketability and market data availability.

[22] Canada, the euro area, India, Japan, Mexico, South Africa, Switzerland, the United Kingdom and the United States.

[23] According to the authors, they are able to do so thanks to their ability to warehouse large amounts of Treasuries, and they have the incentive to do it because that allows them to exploit a regulatory arbitrage opportunity (bundling a 0 risk-weigh Treasury repo with a 100 percent uncollateralized loan would allow them to make a risky loan to counterparties without needing to allocate scarce equity capital).

[24] FSB (2026) also considers the possibility that negative haircuts could be the result of netting at the portfolio level. Julliard et al (2024) examine the drivers of zero haircuts in the UK repo market.

The usual expectation of positive haircuts is based on the parallel between reverse repos and a collateralized loan. Best practices would require that the lender pursues some degree of overcollateralization as a solid mitigant against the borrower's default risk (such as with Loan-to-Value lower than 100 percent in the origination of mortgage loans).

Alternatively, zero or negative haircuts could be justified by a more symmetric view of risks and the search for some balance between the risks incurred by the two sides of the repo transaction. Kahn and McCormick (2025) adopt such an approach in describing a framework in which the haircuts are established with the aim of providing appropriate net protection to both sides of a transaction and reflect both counterparty and collateral risk. There is a fundamental symmetry in the positions of the two participants in a transaction, because "If the borrower in a repo defaults, the lender is left holding securities that may have fallen in value", but, at the same time, "if the lender defaults, the borrower retains the cash but loses the posted securities, which may have risen in value" (ibid.). In acknowledging this symmetry, they propose a way to set haircuts 'proportionately', by taking into consideration equivalent risk measures for the two sides (for example, the 1st and 99th percentile of the probability distribution of collateral returns for the cash lender and borrower, respectively) and weighting the two measures by the PD of the two sides (that of the cash borrower and cash lender, respectively):

$$h = -p \cdot r_{0.01} - q \cdot r_{0.99}$$

where $r_{0.01}$ and $r_{0.99}$ are the 1st and 99th percentile of the collateral's return distribution, p is the PD of the cash borrower and q is the PD of the cash lender. If the (absolute) value of the first term on the right-hand side (i.e. the conditional expected loss of the cash lender) is larger than that of the second term (i.e. the conditional expected loss of the cash borrower), then the resulting haircut should be positive; and vice versa in the opposite case. While the quantitative result is not necessarily indicative of an appropriate level for the haircut,[25] the sign of the result conveys helpful information about which of the two sides needs 'more protection' from a (pure) risk-management perspective.

The 'symmetric' approach is applied to the dataset already used in the previous section.[26] We use the same portfolio, market return, and repo counterpart PDs used in the estimation of NDLs. To estimate '*h*' in the equation above, we should also use CCPs' PDs; these, however, are generally not available; in some rare cases, rating agencies assign credit ratings to some CCPs.[27] In the absence of broad data on CCPs' PDs, we reverse the haircut equation above: we isolate q to back out its hypothetical value when h is equal to zero: this allows us to identify the boundary between positive and negative haircuts. In particular, in this framework positive haircuts are justified as long as:

$$q < -p\frac{r_{0.01}}{r_{0.99}}$$

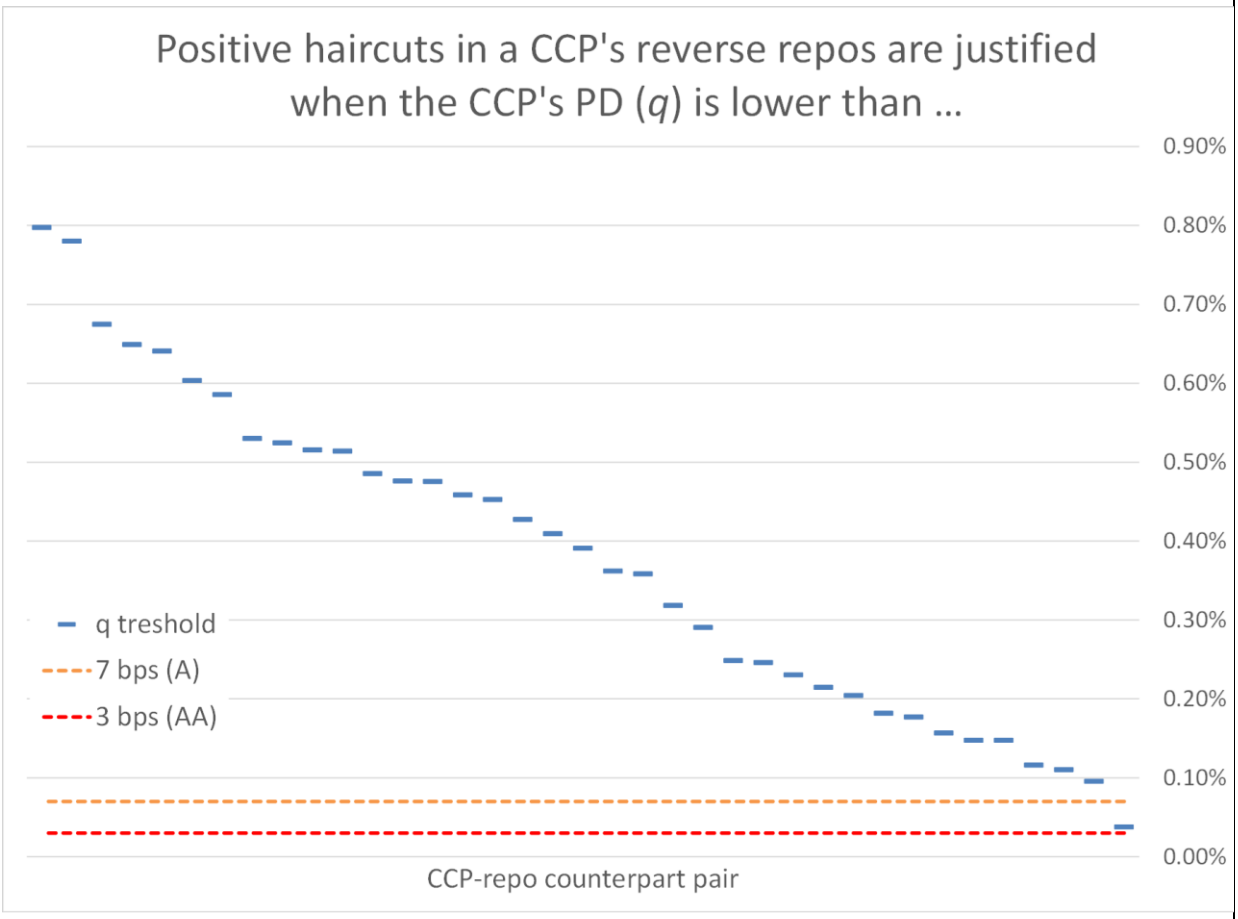
i.e. as long as the PD of a CCP is lower than a given threshold, based on the repo counterpart's PD and the ratio between the left and right tail percentile (at specular confidence level) of the distribution of collateral returns. The results are shown in Figure 5.

[25] The calculation boils down to a comparison of sort of 'conditional expected losses' (conditional on the collateral return hitting either tail of the distribution), while haircuts should be based on the actual loss, in case of the counterpart's default, not the expected loss.

[26] The 1st and 99th percentiles are calculated on the time series of collateral daily returns between 1/1/2020 and 12/30/2024; for p we take the average PDs of each CCP counterpart in the period 2020-2024.

[27] Most EU CCPs are not rated. In the sample used for this work, the two CCPs with a rating are both rated AA-.

Figure 5. Hypothetical exercise on haircuts applied by CCPs to repo collateral



Source: authors, S&P Global

Each blue sign in the chart indicates, for each CCP-repo counterpart pair, the largest CCP PD value that would still justify a positive haircut (i.e. a haircut in favor of the CCP, as cash lender). If the actual PD of the CCP is higher than that value, then a positive haircut is no longer justified (while a negative one is).

In order to get a sense of how those '*q*' thresholds compare to reasonable PD levels for CCPs, we conduct a hypothetical exercise. In the absence of credit ratings for most CCPs in the sample, we (S&P Global 2025, Table 41). The results indicate that in all but one of the cases in the sample a positive haircut would be justified if the CCP were rated A- to A +; equivalently, a negative haircut would be justified only in one case. If the CCPs were rated AA- or higher, the percentage of 'justifiable' positive haircuts would rise to 100 percent (i.e. a negative haircut would never be justified).[28] These figures are to be compared with the 40 percent of reverse repo transactions with actual negative haircuts in the sample.

[28] The average PD associated to the broader AA 'class' of ratings (AA- to AA+) in S&P statistics is 0.03 (3 basis points); the lowest 'q threshold' across CCP-counterpart pairs is 3.8 basis points.

Considering the general market perception of CCPs as safe counterparts, it appears implausible that the application of negative haircuts to the collateral pledged to a CCP in a reverse repo transaction be justifiable in most cases, based on the CCP-repo counterpart pairs in the sample. When this happens nonetheless, it is important that the CCP correctly calculates its prudential requirement for counterparty credit risk, which should compensate, in terms of capital, for the heightened level of risk incurred.[29]

The above approach adopts a pure risk management perspective, abstracting from systemic risk and financial stability considerations. A CCP, however, carries considerable systemic risk, given the profound financial stability implications of its hypothetical default. On the other hand, the same applies to some of its counterparts, which sometimes are also clearing members, and often globally systemic financial institutions: each side of the repo transaction is likely to be Too Big To Fail in its own way (see also Cononico et al (2025)). Ex-ante it is not clear which of the two sides is most likely to act as a stronger risk amplifier in times of stress – and that's why developing comprehensive system-wide stress test exercises involving both CCPs and their clearing members is crucial.[30] The above approach should then be reformulated by focusing less on the 'micro' perspective of which of the two parties (CCPs or their counterparts) is facing a higher conditional expected loss and more on the relative systemic importance of the two.

Apart from the potential destabilization of CCPs' investments, the application of zero or negative haircuts to reverse repo collateral can have further implications for financial stability, such as excessive leverage and procyclicality. Zero or negative haircuts (together with high levels of collateral reuse or rehypothecation) may allow the buildup of leverage (FSB 2026). They can also induce procyclicality: Bejarano and Lu (2026) observe that the US dealer banks which are more likely to apply negative haircuts in their reverse repo transactions during 'normal' (non-stressed) times are also those that more quickly turn to applying positive ones when facing a market crisis (unlike other cash lenders that regularly apply positive haircuts), thus leading to procyclical outcomes that could further amplify the initial shock.

[29] As observed in FSB (2026), "To the extent that transactions with zero haircuts are not adequately covered by other types of risk mitigants, repo collateral may not fully protect lenders in the NCCBR [non-centrally cleared bilateral repo] market segment from credit risk."

[30] The Bank of England's system-wide exploratory scenario (SWES) exercise (BoE 2024) was one of the first to be developed.

Conclusions

Because of the central, systemic role in the financial system assigned to central counterparties (CCPs) by the post-GFC regulatory reforms, they have become pivotal shock-absorbers in modern market structure. Their resilience hinges not only on management of defaults within the clearing waterfall, but also on their ability to manage non-default losses (NDLs) arising outside of it. This paper focuses on the investment channel of NDLs—losses stemming from the management of large cash balances received as margins and default fund contributions and reinvested in deposits, securities, and reverse repos. The analysis highlights that the incentives to lower clearing costs through higher treasury income (generated by the investment of cash received) can also create exposures that are less visible than clearing risks, particularly when investment practices embed reliance on secured funding markets and collateral haircuts are not sufficiently prudent.

We propose a unified, model-based framework to quantify unexpected investment losses by combining historical simulations for market moves with a counterparty-default overlay for reverse repos. For direct investments, bond revaluations are approximated through duration and convexity and mapped to relevant yield-curve dynamics, allowing coherent valuation across currencies and tenors over a five-day horizon. For reverse repos, collateral shocks – based on the same historical simulation component used for direct investments – are integrated with simulated defaults of repo counterparties to derive a loss distribution encompassing all possible default combinations.

Losses are summarized through expected shortfall metrics and benchmarked against proxies for loss-absorbing capacity: CCPs have several tools at their disposal to cover potential NDLs; the available tools can draw on internal or external sources of funds and can vary depending on the CCP's ownership structure (e.g. industry utility vs for-profit ownership). Some tools draw on financial resources held in the form of highly liquid assets; other assets might take time to realize, which can create liquidity gaps. CCPs generally consider own funds, in particular liquid net assets funded by equity, to be the primary resource for covering NDLs. Similar to rule-based loss allocations for default losses, some CCPs include in their rulebooks the potential allocation of losses to clearing members, most often for custody and investment losses. We benchmark the expected shortfall results against skin-in-the-game and the default fund, providing a practical, although indirect, bridge between quantified NDLs and the resources that may ultimately bear them.

The results from the application of our approach to a representative sample of CCPs operating in – or of systematic importance for – the European Union (EU) point to a clear asymmetry across risk channels. Market risk in direct investments appears muted, consistent with strict eligibility and maturity constraints and relatively low portfolio durations for most EU CCPs. By contrast, counterparty credit risk in reverse repo can be material for a subset of CCPs, notably where exposures are concentrated and where collateral protection is weak. This highlights how non-default losses could become a significant source of vulnerability for some CCPs operating in the EU

The combination of low or non-positive haircuts with collateral of significant duration implies that relatively modest market moves can translate into under-collateralization precisely when counterparties are most likely to be under stress. A hypothetical exercise we conduct suggests that non-positive haircuts are rarely consistent with a risk-based calibration when plausible counterparty default probabilities and collateral tail risks are considered. Where market conventions or bargaining outcomes result in haircuts that do not cover potential adverse collateral moves at default, the residual risk must be borne

elsewhere—either through dedicated capital, through loss allocation mechanisms in recovery, or through a combination of both.

Better estimates of non-default losses can support more credible recovery and resolution planning. They can inform when to activate recovery and resolution tools and help assess whether CCPs' own resources are sufficient in times of stress. This is important to limit spillovers from CCPs to the wider clearing ecosystem and to strengthen financial stability.

Avenues for further work

This paper lays the foundation for further work to quantify the risks from CCPs' investments and to assess their relevant recovery provisions. Several extensions follow naturally from the analysis already presented: improving observability of effective haircuts and repo rates; sharpening the mapping between achieved rates and the market (and liquidity) risk of accepted collateral; the extent to which CCPs recycle liquidity with their own clearing members; and integrating quantified NDLs estimates with CCP-specific recovery provisions to assess how losses would be transmitted to clearing members under stress. Our work also connects to the ongoing debate on greater centralization of repo clearing: if CCPs were able to place more of their repo activity into centrally cleared repos, their reverse-repo exposures would more naturally benefit from CCP risk management and mutualization features, reducing bilateral counterparty credit risk. Broadening our approach to incorporate additional NDLs channels (beyond investments) would further strengthen its usefulness for comprehensive resilience assessments. Overall, richer and more comparable estimates of investment-related NDLs can enhance the credibility of recovery and resolution planning by clarifying when tools may be activated, how large the resource needs could be, and how spillovers to the wider clearing ecosystem might propagate during market downturns.

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